Group I, Row V: N_2

The setup for the cross section:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.1}$$

This can be written as:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c U^{af}(y)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.2}$$

This term contains a UV divergence at $x\to z$. So, we use the following identity to isolate the UV term and the finite term. The identity as follows:

$$f^{abc}\,f^{def}\,U^{be}(z)\,U^{cf}(z)=N_c\,U^{ad}(z)$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.3}$$

This can be written in two parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.4}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.5}$$

We will handle the p^+, using + prescription for the first part i.e., $2a$. We get the following:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{0.6}$$

Now, lets go ahead with the second part which has UV divergence. So, we will use dimensional regularization to handle it. We will write it in two parts:

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2b\,\mathrm{UV}}}{d^3k}+\frac{d^3N_{2b\,\mathrm{rest}}}{d^3k} \tag{0.7}$$

Following the UV subtraction with the exponential and the energy denominator evaluated at $z=x$, the subtraction term (which has exactly the $z\to x$ singularity of N_{2b}) is

$$\frac{d^3N_{2b\,\mathrm{UV}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]$$
$$\times\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y). \tag{0.8}$$

The z integral is done with dimensional regularisation in $d=2-\epsilon$ ($\overline{MS}$), using the master integral with the trace taken as $\delta^{mm}=d$:

$$\int_z\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}=\int_z\frac{1}{(z-x)^2}-\delta^{mm'}\int_z\frac{(z-x)^m\,(z-y)^{m'}}{(z-x)^2\,(z-y)^2}=0+\pi\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-x)^2\,\bar{\mu}^2_{\overline{MS}}}{4}\right], \tag{0.9}$$

where the first integral is scaleless ($\int d^2p/p^2=0$). The p^+ integral is exact:

$$\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}dp^+\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]=2\log\left(\frac{k^+-\Lambda}{\Lambda}\right)-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}. \tag{0.10}$$

Hence

$$\frac{d^3N_{2b\,\mathrm{UV}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{k^+-\Lambda}{\Lambda}\right) - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\right]\int_{x,y,x',w'} e^{-ik\cdot(w'-x)}$$
$$\times\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-x)^2\,\bar{\mu}^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \qquad (0.11)$$

Setting $\Lambda \to 0$, where ever possible

$$\frac{d^3N_{2b\,\mathrm{UV}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{k^+}{\Lambda}\right) - \frac{11}{6}\right]\int_{x,y,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-x)^2\,\bar{\mu}^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}$$
$$\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \qquad (0.12)$$

The remainder is the full integrand of N_{2b} minus the subtraction term; it is UV finite and is computed in $d = 2$ ($d_\perp = 2$). Both lines must be kept under the same $\int_z$:

$$\frac{d^3N_{2b\,\mathrm{rest}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}\Bigg\{e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{p^+\,(y-x)^2 + (k^+-p^+)\,(y-z)^2}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2\,(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]$$
$$-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2} + \frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right] + \frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2\,(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2} - \frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2} + \frac{(y-z)^m\,(x-z)^i}{(x-z)^2} + \frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$-\,e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}\Bigg\}$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \qquad (0.13)$$

Using plus prescription to isolate the p^+ pole and setting $\Lambda \to 0$, wherever possible

$$
\begin{aligned}
\frac{d^3 N_{2b\,\mathrm{rest}}}{d^3 k} = {} & \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}\Bigg\{ e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
& \times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}} - \left[\frac{k^+}{k^+-p^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right] \\
& + \left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right] \\
& - \left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right] \\
& + \frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right] \\
& - \left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg] \\
& - e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}-2\right]\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}\Bigg\} \\
& \times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
& - \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{N_c}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right. \\
& \left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2\,(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
& - \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{N_c}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right. \\
& \left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2\,(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
& - \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{N_c}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2} \\
& \times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y). \qquad (0.14)
\end{aligned}
$$

Remarks.

- $N_2 = N_{2a} + N_{2b\,\mathrm{UV}} + N_{2b\,\mathrm{rest}}$ exactly. The only $\log\bar{\mu}$ is in $N_{2b\,\mathrm{UV}}$. In the convention $\int d^2p/p^2 = 0$ the $\frac{2}{\epsilon}$ is dropped.
- The three $\log(k^+/\Lambda)$ terms of $N_{2b\,\mathrm{rest}}$ are UV finite as a set: at $z \to x$ they behave as $(-1-1+2)\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2(x-z)^2} = 0$. The Λ-independent part of $N_{2b\,\mathrm{rest}}$ is UV finite as well: there the $[\cdot]_+$ terms have zero UV coefficient, and the regular coefficients $\frac{p^+(k^+-p^+)}{k^{+2}} - 2$ cancel against the subtracted line.
- The $\log(k^+/\Lambda)$ terms of N_{2a} and $N_{2b\,\mathrm{rest}}$ go to the JIMWLK terms. The $-\frac{11}{6}\log\bar{\mu}^2$ of $N_{2b\,\mathrm{UV}}$ is the running-coupling term, which combines with Group I Row II.